From Geometric Carrier to Physical Carrier: Distinguishability, Holonomy and the Actualisation of Information in Compact S^3 Geometry

Andrei Preece and Boris Batenin
Independent Researchers,
Berlin, 2026

ABSTRACT

The Holonomic Compact Cosmology programme has established that compact S^3 geometry supports a finite causal coherence structure, and that expansion of the causal 3-sphere requires internal reconstruction of distinguishable states. The present paper asks a prior question: what transforms a geometric carrier into a physical carrier? We argue that geometry alone is not yet a carrier. A carrier becomes physical only when distinguishability can be realised and preserved under the structural constraints of the geometry. We investigate whether holonomy—the geometric phase acquired by a quantum state under parallel transport—provides a more fundamental structural principle for admissible distinguishability than entropy ceilings alone. The paper traces a logical chain from S^0 (minimal distinction) through spinor representation, the Hopf fibration, the Bloch sphere, and Berry phase, to the emergence of physically realised distinguishability. We do not claim to derive the Bekenstein bound or to replace holography. We ask whether a quantum-geometric mechanism can explain why physically admissible distinguishability is limited before one introduces entropy bounds. The work concludes by proposing a long-term research programme rather than a closed solution.

Keywords: distinguishability, holonomy, Berry phase, Hopf fibration, spinor, geometric carrier, physical carrier, information, compact S^3.

1. INTRODUCTION: THE QUESTION BEFORE RECONSTRUCTION

The HCC programme has established two structural results concerning compact S^3 geometry.

First, light functions as a causal coherence structure. On S^3_R, null propagation defines a finite causal accessibility timescale $t_coh = \pi R/c$, and the causal accessibility ratio $\Gamma(T)$ reaches unity at finite time (Preece and Batenin, 2026a). The speed of light c is a causal self-consistency parameter, not an arbitrary empirical input.

Second, expansion requires reconstruction. If the causal 3-sphere expands according to $R_N = \ell_P \varphi^N$, then reconstruction times must co-scale as $t_N = t_P \varphi^N$. The newly accessible causal volume must acquire internal distinguishability through a physical mechanism. Fractal re-quantisation was proposed as a candidate (Preece and Batenin, 2026c).

These results assume that a geometric carrier is available and ask what must happen for it to expand consistently.

The present paper asks a prior question.

What makes a geometric carrier become a physical carrier in the first place?

Geometry defines a topological possibility. A compact S^3 is a manifold. It has points, geodesics, a metric, a Laplacian, a spectrum. But a manifold of points is not yet a physical arena. Points in a manifold are not distinguishable in the quantum sense. They are labels, not states. A geometric carrier must become a physical carrier: a structure in which distinguishable states can be realised, preserved, and transformed under unitary evolution.

The central thesis of this paper is:

Geometry alone is not yet a carrier. A carrier becomes physical only when distinguishability can be realised and preserved under the structural constraints of the geometry.

This thesis redirects attention from the quantity of information (how many bits can be stored) to the condition of distinguishability (what makes a bit possible at all). Information is secondary. Distinguishability is primary. Information exists only because distinguishability exists.

We investigate whether holonomy—the geometric phase acquired by a quantum state under parallel transport in a curved space—provides a more fundamental structural principle for admissible distinguishability than entropy ceilings alone.

2. THE DISTINCTION BETWEEN GEOMETRIC AND PHYSICAL CARRIERS

2.1 Definitions

Definition (Geometric carrier). A geometric carrier is a smooth manifold M equipped with a metric g, a causal structure, and a topology. It provides the formal possibility of points, paths, distances, and regions. It is a mathematical object.

Definition (Physical carrier). A physical carrier is a geometric carrier in which distinguishable quantum states can be realised, preserved under transport, and transformed by unitary operations. It is a geometric carrier that has become an arena for physics.

The gap between them is not formal but structural. A geometric carrier becomes physical when the following conditions are met:

- Distinguishability: there exist orthogonal states that can be prepared and distinguished by local operations.
- Stability: the phase relations that define these states are preserved under the geometric transport that the manifold allows.
- Unitary evolution: the transformations generated by transport in the manifold form a unitary representation.

2.2 What Geometry Does Not Provide

A manifold of points does not, by itself, distinguish its points in a quantum-mechanically meaningful way. The points are a continuum. Any two points are connected by a continuous path. The Hilbert space of a particle on a manifold is infinite-dimensional, but the position eigenstates are not normalisable and not physically realisable as distinguishable states. They are idealisations.

To obtain physically distinguishable states, one needs more than a manifold. One needs a structure that:

- Selects a finite set of orthogonal directions at each point (a fibre).
- Specifies how these directions transform under transport around closed loops (holonomy).
- Ensures that the resulting phases are stable and observable (Berry phase).

Geometry supplies the stage. Distinguishability supplies the actors. Holonomy is the script that tells them how to move.

3. THE LOGICAL CHAIN: FROM S^0 TO HOLONOMY

We trace a conceptual chain from the minimal condition of distinction to the emergence of physically realised distinguishability. This chain is not a derivation. It is an exploration of structural dependencies. Each step introduces a new condition that the previous step does not supply.

3.1 S^0: Minimal Distinction

S^0 is the zero-dimensional sphere: a pair of points.

Structural role. S^0 is the minimal condition of distinction. It has two elements and no internal structure. It is the logical ancestor of the bit.

What it lacks. S^0 has no connection. The two points are distinct but unrelated. There is no path between them, no way to transform one into the other, no notion of phase or superposition. Distinction without connection is not yet a physical state space.

3.2 From S^0 to Spinor Representation

To connect the two points of S^0, one passes to a continuous space that contains them as a boundary or as antipodal points. The simplest such space is the circle S^1, but the relevant connection for quantum mechanics is the passage to the two-dimensional complex vector space.

Spinor representation. A two-state quantum system is described by a vector in $\mathbb{C}^2$. The two basis states $|0\rangle$ and $|1\rangle$ are the quantum descendants of the two points of S^0. But now they are connected: any superposition $\alpha|0\rangle + \beta|1\rangle$ is a valid state. The sphere of normalised states is S^3, the three-sphere.

Definition (Spinor). A spinor is an element of a complex vector space that carries a projective representation of the rotation group. Under a rotation by 2π, a spinor changes sign. It requires a rotation by 4π to return to its original value.

Structural role. The spinor representation introduces the first genuinely quantum element: the relative phase between basis states. The two points of S^0 become the two hemispheres of a continuous quantum state space. Distinction is no longer binary; it is a matter of degree, encoded in the complex amplitude.

3.3 The Hopf Fibration

The space of normalised spinor states is S^3. This space has a remarkable geometric structure: the Hopf fibration.

Proposition (Hopf fibration). The three-sphere S^3 can be fibred over the two-sphere S^2 with fibre S^1:

$$S^1 \hookrightarrow S^3 \to S^2.$$

Explicitly, a point $(z_1, z_2) \in S^3 \subset \mathbb{C}^2$ with $|z_1|^2 + |z_2|^2 = 1$ maps to the point on the Bloch sphere with coordinates:

$$n = (2\,\mathrm{Re}(z_1^* z_2),\ 2\,\mathrm{Im}(z_1^* z_2),\ |z_1|^2 - |z_2|^2).$$

The fibre over each point is a circle S^1, representing the overall phase of the state.

Structural role. The Hopf fibration separates the quantum state into two components:

- The base S^2 (the Bloch sphere) encodes the relative amplitude and relative phase between the basis states—the physically observable distinguishability.
- The fibre S^1 encodes the overall phase—the unobservable but geometrically crucial degree of freedom.

This separation is not an arbitrary mathematical convenience. It is a geometric fact about the state space of a two-level quantum system. The base and fibre are not independent; they are woven together by the fibration. The twisting of the fibre over the base is measured by the Hopf invariant, which is unity for this fibration.

3.4 The Bloch Sphere

The Bloch sphere is the base space S^2 of the Hopf fibration. Every point on the Bloch sphere corresponds to a physically distinct pure state of a two-level system. The north and south poles are orthogonal states. Points on the equator are equal superpositions with different relative phases.

Definition (Bloch sphere distinguishability). Two pure states are physically distinguishable if and only if they correspond to distinct points on the Bloch sphere. States that differ only by an overall phase (i.e., that lie on the same Hopf fibre) are not physically distinguishable.

Structural role. The Bloch sphere is the first complete example of a physical carrier. It is a geometric object (the sphere S^2) whose points correspond to physically distinguishable states. The geometry of the sphere—its distances, angles, and areas—directly encodes the statistical distinguishability of quantum states. The Fubini-Study metric on the Bloch sphere gives the distance between two pure states as the angle between their Bloch vectors.

3.5 Berry Phase and Holonomy

A quantum state transported around a closed path in parameter space acquires a geometric phase in addition to the dynamical phase. This is the Berry phase.

Definition (Berry phase). Let $|\psi(t)\rangle$ be a state evolving under a time-dependent Hamiltonian $H(t)$ such that $H(T) = H(0)$. If the evolution is adiabatic, the state at time T is:

$$|\psi(T)\rangle = \exp(i\gamma)\,\exp\left(-(i/\hbar)\int E(t)\,dt\right)|\psi(0)\rangle,$$

where the first factor is the geometric Berry phase:

$$\gamma = i \oint \langle\psi|\nabla\psi\rangle \cdot d\lambda,$$

and the second is the dynamical phase. The Berry phase depends only on the path in parameter space, not on the speed of traversal.

Structural role. The Berry phase is a holonomy. It measures the global geometric twist of the quantum state space over the parameter space. It reveals that the phase of a quantum state is not an arbitrary gauge freedom but carries physical information about the geometry of the state space.

Definition (Holonomy). Holonomy is the geometric transformation acquired by a vector or state under parallel transport around a closed loop. In a space with curvature or twist, parallel transport around a loop does not return the state to its original value. The failure to close is the holonomy.

Examples of holonomy in physics:

- Berry phase in quantum mechanics.
- Aharonov-Bohm effect (holonomy of the electromagnetic connection).
- Wilczek-Zee phase for degenerate states.
- Gravitational holonomy around cosmic strings.

Holonomy is a fundamental structural principle. It is not specific to any particular interaction. It arises whenever a state is transported in a space with non-trivial geometry.

4. HOLONOMY AS A CONSTRAINT ON DISTINGUISHABILITY

4.1 The Question

The Bekenstein bound limits the amount of information that can be contained in a region of space: $I \leq 2\pi RE/(\hbar c \ln 2)$. This is an entropy ceiling. It is a kinematic bound derived from general principles of thermodynamics and black hole physics.

We ask a different question. Before one asks how much information can be stored, one must ask what makes information possible. Information requires distinguishability. Distinguishability requires that there exist orthogonal states that can be prepared, preserved, and distinguished by local measurements.

What limits the existence of such states?

4.2 A Structural Conjecture

Conjecture (Holonomy as a distinguishability constraint). In a compact curved geometry, the admissibility of physically distinguishable states is constrained not only by entropy bounds but by the holonomy structure of the geometry. States that cannot be stably transported around closed loops without losing their phase relations cannot serve as carriers of physical distinguishability.

This is not a replacement for the Bekenstein bound. It is a different kind of constraint, operating at the level of state structure rather than state counting.

4.3 Illustrative Argument

Consider a quantum state defined on a compact S^3. The state is transported around a closed geodesic loop. After one circuit, it acquires a geometric phase γ determined by the curvature enclosed by the loop and the connection on the state space.

If the phase γ is not stable under small perturbations of the path, the state cannot serve as a reliable carrier of distinguishability. The phase is part of what defines the state. An unstable phase means an unstable state.

Now consider a large number of states that an observer might wish to distinguish. These states must be transported along different paths through the geometry as the observer moves and as the system evolves. The relative phases between the states must remain stable under this transport; otherwise, the states will become mixed and lose their mutual distinguishability.

Proposition (Phase stability condition). A set of states on a geometric carrier can serve as a physical basis of distinguishable states only if the holonomy group of the carrier acts on the set in a way that preserves the relative phases, up to a globally defined gauge transformation.

This condition is automatically satisfied in flat space with trivial holonomy. In curved space, it is a non-trivial constraint. It selects, from the space of all possible quantum states, those that are geometrically stable.

4.4 Distinguishability as a Geometric Resource

If holonomy constrains distinguishability, then distinguishability is not a free resource. It is paid for by the geometric structure of the carrier. The question "how many distinguishable states can be realised?" becomes a question about the holonomy of the geometry, not merely about its volume.

Open Problem 1. For a compact S^3 carrying a specified connection, determine the maximum number of mutually distinguishable pure states that can be stably transported around all closed geodesic loops up to length L.

Open Problem 2. Determine the relationship between the holonomy constraint on distinguishability and the Bekenstein entropy bound. Are they independent? Does one imply the other? Or does holonomy provide a microscopic mechanism that enforces the entropy bound?

We do not answer these questions here. We identify them as the core of a research programme.

5. FROM HOLONOMY TO PHYSICAL CARRIER

5.1 Conditions for a Physical Carrier

We now summarise the conditions that transform a geometric carrier into a physical carrier.

Axiom (Physical carrier conditions). A geometric carrier M becomes a physical carrier if and only if:

1. **Topological possibility.** M is a smooth manifold with a metric and a causal structure. (Geometric carrier condition.)
2. **Distinguishability.** There exists a Hilbert space of states on M such that orthogonal states can be prepared and distinguished by local operations.
3. **Phase stability.** The relative phases that define the distinguishable states are preserved under parallel transport around all closed paths in M that correspond to physically accessible operations.
4. **Unitary evolution.** The transport operators form a unitary representation of the holonomy group of M on the state space.

Conditions 2–4 are not guaranteed by Condition 1. They require additional structure: a fibre bundle over M, a connection on that bundle, and a stability property of the connection.

The Hopf fibration provides the minimal example. S^3 is the total space. S^2 is the base space of physically distinguishable states. S^1 is the fibre of unobservable phases. The Berry phase is the holonomy of the connection on this bundle. The Bloch sphere is the physical carrier that results.

5.2 Information as Secondary

In this framework, information is not fundamental. It is a measure of distinguishability that has already been realised.

Definition (Information, operational). Information is the capacity to encode and discriminate among a set of mutually distinguishable states that have been physically realised on a carrier.

One cannot speak of information before one has a physical carrier. The Bekenstein bound counts how many bits can be stored in a region, assuming that distinguishable states can be realised up to the limit set by the bound. Our question is whether the bound can always be saturated, or whether geometric constraints—holonomy, curvature, topology—may impose stricter limits on which states are physically admissible, even if the entropy budget would formally allow more.

Conjecture (Holonomic distinguishability bound). On a compact curved geometry, the number of physically realisable distinguishable states N_dist satisfies:

$$N_dist \leq \min(N_Bekenstein, N_holonomy),$$

where N_Bekenstein is the entropy bound and N_holonomy is a geometric bound determined by the holonomy structure of the carrier.

If N_holonomy is strictly smaller than N_Bekenstein for certain geometries, then holonomy provides an independent, more fundamental limit on physical information.

6. POSSIBLE PHYSICAL CONSEQUENCES

The following are speculative consequences that would follow if the holonomic distinguishability bound can be established. They are presented as conjectures, not as results.

Conjecture (Vacuum distinguishability and Λ). The vacuum energy density of a compact geometry may be related not to the total number of Planck cells in the volume, but to the number of holonomically stable distinguishable states that the geometry can support. If the holonomic bound is stricter than the volumetric bound, this could provide a new perspective on the magnitude of the cosmological constant.

Conjecture (Dimensional reduction). At scales where the holonomy group becomes non-trivial, the effective number of distinguishable states may scale with a lower effective dimension than the geometric dimension. This could connect to the observed dimensional reduction in quantum gravity approaches.

Conjecture (Black hole information). The holonomic distinguishability bound may provide a mechanism for the finiteness of black hole entropy that does not rely solely on the Bekenstein-Hawking area law. The horizon geometry may support only a finite number of holonomically stable states, with the area scaling emerging as a consequence rather than a postulate.

These are directions for investigation, not claims.

7. A LONG-TERM RESEARCH PROGRAMME

We propose a research programme organised around five themes.

Theme 1: Holonomy Groups of Compact S^3

Classify the holonomy groups that can arise on compact S^3 with physically motivated connections (spin connection, electromagnetic, Yang-Mills). Determine the invariant subspaces of the state space under the action of each holonomy group.

Theme 2: Phase Stability Under Geometric Transport

Define a quantitative measure of phase stability for states transported around closed loops in a curved geometry. Determine the conditions under which a set of states remains mutually distinguishable after transport through a given set of loops.

Theme 3: Holonomic Distinguishability Bound

Prove or disprove the conjecture that the number of physically distinguishable states on a compact geometry is bounded by a function of the holonomy group, independent of the volumetric Bekenstein bound. If proved, compute this function for S^3 with the spin connection.

Theme 4: Relation to the Cosmological Constant

If Theme 3 yields a holonomic bound on vacuum distinguishability, compute the associated vacuum energy density for a compact S^3 of radius R. Compare with the observed Λ_{obs} and determine whether the bound can accommodate the observed value.

Theme 5: Connection to the HCC Reconstruction Programme

Integrate the holonomic distinguishability constraint into the fractal re-quantisation mechanism of Preece and Batenin (2026c). At each reconstruction step $N \to N+1$, the number of new distinguishable states generated should satisfy the holonomic bound. Determine whether this modifies the φ-scaling of the reconstruction hierarchy.

8. SUMMARY OF DEFINITIONS, AXIOMS, PROPOSITIONS, AND CONJECTURES

For clarity, we collect the formal statements introduced in this paper.

Definition (Geometric carrier). A smooth manifold M with metric g, causal structure, and topology.

Definition (Physical carrier). A geometric carrier in which distinguishable quantum states can be realised, preserved under transport, and transformed by unitary operations.

Definition (Holonomy). The geometric transformation acquired by a state under parallel transport around a closed loop.

Definition (Berry phase). The geometric phase acquired by an adiabatically transported quantum state, given by

$$\gamma = i \oint \langle \psi | \nabla \psi \rangle \cdot d\lambda.$$

Definition (Information, operational). The capacity to encode and discriminate among a set of mutually distinguishable states on a physical carrier.

Axiom (Physical carrier conditions). A geometric carrier becomes physical when it supports distinguishability, phase stability, and unitary evolution under its holonomy group.

Proposition (Phase stability condition). A set of states can serve as a physical basis only if the holonomy group preserves their relative phases up to a global gauge transformation.

Conjecture (Holonomic distinguishability bound). $N_{dist} \leq \min(N_{Bekenstein}, N_{holonomy})$.

Conjecture (Vacuum distinguishability and Λ). Λ_{eff} is constrained by the holonomic distinguishability bound on the vacuum geometry.

Open Problem 1. Maximum number of holonomically stable distinguishable states on compact S^3.

Open Problem 2. Relationship between holonomic and Bekenstein bounds.

Open Problem 3. Computation of Λ from the holonomic distinguishability bound.

CONCLUSION: THE MISSING BRIDGE

The HCC programme has established the geometry of compact S^3 as a causal coherence structure. It has shown that expansion requires reconstruction. These are results about the geometric carrier.

The present work asks what transforms a geometric carrier into a physical carrier. We have proposed that holonomy is the structural principle that bridges geometry and distinguishability. The Berry phase, the Hopf fibration, and the spinor representation are not isolated phenomena; they are elements of a coherent geometric language for how physical states arise from topological possibility.

The central chain is:

$S^0 \to$ minimal distinction

$\to$ spinor representation

$\to$ Hopf fibration

$\to$ Bloch sphere

$\to$ Berry phase

$\to$ holonomy

$\to$ physical carrier

$\to$ physically realised distinguishability

$\to$ information.

Information is not the starting point. It is the consequence. Before one counts bits, one must understand what makes a bit possible.

This work does not claim to have solved the problem. It claims to have identified the problem. The long-term research programme outlined in Section 7 is offered as a path forward. If successful, it may provide the missing bridge between topology, quantum geometry, distinguishability, and information that the HCC programme requires to complete its physical interpretation.

The previous HCC papers establish the geometry. The present work asks what transforms geometry into a physical carrier. This may become the missing bridge.

REFERENCES

[1] M. V. Berry, "Quantal Phase Factors Accompanying Adiabatic Changes," Proceedings of the Royal Society A, 392, 45–57, 1984.

[2] A. Shapere and F. Wilczek, Geometric Phases in Physics, World Scientific, 1989.

[3] B. Simon, "Holonomy, the Quantum Adiabatic Theorem, and Berry's Phase," Physical Review Letters, 51, 2167–2170, 1983.

[4] H. Hopf, "Über die Abbildungen der dreidimensionalen Sphäre auf die Kugelfläche," Mathematische Annalen, 104, 637–665, 1931.

[5] F. Bloch, "Nuclear Induction," Physical Review, 70, 460–474, 1946.

[6] J. J. Sakurai and J. Napolitano, Modern Quantum Mechanics, 2nd ed., Cambridge University Press, 2017.

[7] M. Nakahara, Geometry, Topology and Physics, 2nd ed., CRC Press, 2003.

[8] J. M. Lee, Introduction to Smooth Manifolds, 2nd ed., Springer, 2013.

[9] B. O'Neill, Semi-Riemannian Geometry with Applications to Relativity, Academic Press, 1983.

[10] M. Spivak, A Comprehensive Introduction to Differential Geometry, Vol. 2, Publish or Perish, 1999.

[11] Y. Aharonov and D. Bohm, "Significance of Electromagnetic Potentials in the Quantum Theory," Physical Review, 115, 485–491, 1959.

[12] F. Wilczek and A. Zee, "Appearance of Gauge Structure in Simple Dynamical Systems," Physical Review Letters, 52, 2111–2114, 1984.

[13] J. D. Bekenstein, "Black Holes and Entropy," Physical Review D, 7, 2333–2346, 1973.

[14] J. D. Bekenstein, "Universal Upper Bound on the Entropy-to-Energy Ratio for Bounded Systems," Physical Review D, 23, 287–298, 1981.

[15] G. 't Hooft, "Dimensional Reduction in Quantum Gravity," arXiv:gr-qc/9310026, 1993.

[16] L. Susskind, "The World as a Hologram," Journal of Mathematical Physics, 36, 6377–6396, 1995.

[17] R. Bousso, "The Holographic Principle," Reviews of Modern Physics, 74, 825–874, 2002.

[18] C. E. Shannon, "A Mathematical Theory of Communication," Bell System Technical Journal, 27, 379–423 and 623–656, 1948.

[19] Preece, A. and Batenin, B. Light as Causal Coherence Structure in Compact S^3 Geometry. HCC Research Branch, Berlin, 2026.

[20] Preece, A. and Batenin, B. Causal Co-Scaling of Distance and Time in a Fractal S^3 Reconstruction Model. Research Note, 2026.

[21] Preece, A. and Batenin, B. Fractal Re-Quantisation as Causal Reconstruction in Compact S^3 Geometry. HCC Research Branch, 2026.